

UBC ELEC 221: Introduction to Signals and Systems

Continuous Time Signals

Even and Odd Components

$$x_{\text{even}}(t) = \frac{1}{2}[x(t) + x(-t)]$$

$$x_{\text{odd}}(t) = \frac{1}{2}[x(t) - x(-t)]$$

Periodicity

$$x(t + kT_0) = x(t) \forall t \in (-\infty, \infty)$$

Energy

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

Power

$$P = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

Causality:

- Causal: $x(t) = 0$ for $t < 0$.
- Anti-Causal: $x(t) = 0$ for $t \geq 0$.
- Acausal or Non-Causal: Both of the above.

Continuous Time Systems

Linearity

$$\mathcal{S}[\alpha x(t) + \beta y(t)] = \alpha \mathcal{S}[x(t)] + \beta \mathcal{S}[y(t)]$$

Time Invariance

$$\mathcal{S}[x(t)] = y(t) \Rightarrow \mathcal{S}[x(t \pm \tau)] = y(t \pm \tau)$$

Zero-State Response. Due to the input as the initial conditions are zero.

Zero-Input Response. Due to the initial conditions as the input is zero.

Convolution

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau = \int_{-\infty}^{\infty} x(t - \tau)h(\tau) d\tau$$

Causality. A continuous time system $\mathcal{S}$ is causal if whenever $x(t) = 0$ and there are no initial conditions, $y(t) = 0$ and the output $y(t)$ does not depend on future inputs. **BIBO Stability.** If an input bounded then the output of a BIBO stable system is bounded.

$$\Leftrightarrow \int_{-\infty}^{\infty} |h(t)| dt < \infty$$

Laplace Transform

Eigenfunction Property

$$\mathcal{S}[e^{s_0 t}] = H(s_0)e^{s_0 t}$$

One Sided Laplace Transform:

$$F(s) = \mathcal{L}[f(t)u(t)] = \int_{0^-}^{\infty} f(t)e^{-st} dt, \forall s = \sigma + j\omega$$

Note: $|H(s)|$ is greatest when s is closest to the poles.

Signal Support

Finite support

Causal function

Anti-causal

Non-causal

ROC

Entire s-plane.

$\sigma > \max(\sigma_i), -\infty < \omega < \infty$

$\sigma < \min(\sigma_i), -\infty < \omega < \infty$

$\mathcal{R} = \mathcal{R}_{\text{causal}} \cap \mathcal{R}_{\text{anti-causal}}$

Initial Value Theorem

$$f(0^+) = \lim_{s \rightarrow \infty} sF(s)$$

Final Value Theorem

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$$

BIBO Stability. If the region of convergence contains the $j\omega$ -axis then the system is BIBO stable.

2-Sided LT from 1-Sided LT

- For an anti-causal function $C(s) = \mathcal{L}[c(-t)u(t)]_{(-s)}$

- For a 2-sided function

$$p(t) = p_{ac}(t) + p_c(t) = p(t)u(-t) + p(t)u(t)$$

- $P(t) = \mathcal{L}[p_{ac}(-t)u(t)]_{(-s)} + \mathcal{L}[p_c(t)u(t)]$

Fourier Series

Eigenfunction Property

$$\mathcal{S}[e^{j\omega_0 t}] = H(j\omega_0)e^{j\omega_0 t}$$

$$x(t) = \sum_k X_k e^{j\omega_k t}$$

$$\Rightarrow y(t) = \sum_k X_k H(j\omega_k) e^{j\omega_k t}$$

$$= \sum_k X_k |H(j\omega_k)| e^{j(\omega_k t + \angle H(\omega_k))}$$

Fourier Series Coefficients (for any t_0)

$$X_k = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} x(t) e^{-jk\omega_0 t} dt$$

Parseval's Power Relation (for any t_0):

$$P = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |X_k|^2$$

Symmetry of Line Spectra:

$$|X_k| = |X_{-k}|$$

$$\angle X_k = -\angle X_{-k}$$

Trigonometric Fourier Series:

$$x(t) = X_0 + 2 \sum_{k=1}^{\infty} |X_k| \cos(k\omega t + \Theta_k)$$

$$x(t) = c_0 + 2 \sum_{k=1}^{\infty} [c_k \cos(k\omega_0 t) + d_k \sin(k\omega_0 t)]$$

$$c_k = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} x(t) \cos(k\omega_0 t) dt \quad k = 0, 1, 2, \dots$$

$$d_k = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} x(t) \sin(k\omega_0 t) dt \quad k = 1, 2, 3, \dots$$

$$\Theta_k = -\arctan(d_k/c_k)$$

Fourier Coefficients from Laplace Transform. If $x_1(t)$ is a single period of $x(t)$, then

$$X_k = \frac{1}{T_0} \mathcal{L}[x_1(t)] \Big|_{s=jk\omega_0}$$

Response of LTI Systems to Periodic Signals. If the input to an LTI system has Fourier Series

$x(t) = X_0 + 2 \sum_{k=1}^{\infty} |X_k| \cos(k\omega t + \angle X_k)$, then the steady state response is

$$y(t) = X_0 |H(j0)| + 2 \sum_{k=1}^{\infty} |X_k| |H(jk\omega_0)| \cos(k\omega_0 t + \angle X_k + \angle H(jk\omega_0))$$

Fourier Transform

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Inverse Fourier Transform

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

Fourier Transform from Laplace Transform (if $X(s)$ contains the $j\omega$ -axis),

$$\mathcal{F}[x(t)] = \mathcal{L}[x(t)] \Big|_{s=j\omega} = X(s) \Big|_{s=j\omega}$$

Fourier Transform of Periodic Signals under Fourier Series

$$\sum_k X_k e^{jk\omega_0 t} \Leftrightarrow \sum_k 2\pi X_k \delta(\omega - k\omega_0)$$

Parseval's Energy Relation

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

Sampling Theory

$$x_s(t) = x(nT_s) = x(t)|_{t=nT_s} = x(t) \sum_n \delta(t - nT_s)$$

$$X_s(\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(\omega - k\omega_s)$$

Nyquist-Shannon Sampling Rate

$$\omega_s = \frac{2\pi}{T_s} \geq 2\omega_{\max}$$

Aliasing occurs if $\omega_s < 2\omega_{\max}$

Reconstruction $X(\omega) = X_s(\omega)H_{\text{lp}}(\omega)$:

$$H_{\text{lp}}(\omega) = \begin{cases} T_s & -\frac{\omega_s}{2} \leq \omega \leq \frac{\omega_s}{2} \\ 0 & \text{otherwise} \end{cases}$$

Inclusive bounds. If represented by a unit step function the bounds are inclusive for filters

Signal Reconstruction from Sinc Interpolation

$$x_r(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \frac{\sin(\pi(t - nT_s)/T_s)}{\pi(t - nT_s)/T_s}$$

Discrete Time Signals

Definition $x[n] = x(nT_0)$.

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n - k]$$

Periodicity

$$x[n + kN] = x[n] \quad \forall k \in \mathbb{Z}$$

When sampling an analog sinusoid of fundamental period T_0 , we obtain a periodic discrete sinusoid provided that m, N not divisible by each-other,

$$\frac{T_s}{T_0} = \frac{m}{N}$$

Aliasing occurs if

$$T_s > T_0/2$$

Energy

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

Power

$$P_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

Convolution

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k]$$

Bounded-Input Bounded-Output Stability

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

Z-Transform

$$\mathcal{Z}[x(nT_s)] = \mathcal{L}[x_s(t)]|_{z=e^{sT_s}} = \sum_n x(nT_s)z^{-n}$$

Convergence

$$|X(z)| = \sum_n |x[n]|r^{-n} < \infty$$

Initial Value Theorem

$$x[0] = \lim_{z \rightarrow \infty} X(z)$$

Final Value Theorem:

$$\lim_{n \rightarrow \infty} x[n] = \lim_{z \rightarrow 1} (z - 1)X(z)$$

BIBO Stability. If the ROC contains radius $z = 1$, then the system is BIBO stable.

Basic Properties of Laplace Transforms

Property	t-domain	s-domain
Linearity	$\alpha f(t) + \beta g(t)$	$\alpha F(s) + \beta G(s)$
1st Derivative	$f'(t)$	$sF(s) - f(0^-)$
2nd Derivative	$f''(t)$	$s^2F(s) - sf(0^-) - f'(0^-)$
Frequency Differentiation	$(-t)^n f(t)$	$\frac{d^n}{ds^n} F(s)$
Integration	$\int_{0^-}^t f(\tau) d\tau$	$\frac{F(s)}{s}$
Frequency Integration	$\frac{f(t)}{t}$	$\int_{-\infty}^{-s} F(-\tau) d\tau$
Convolution	$\int_0^t f(t - \tau)g(\tau) d\tau$	$G(s)F(s)$
Time-shifting	$f(t - \tau)$	$e^{-\tau s}F(s)$
Frequency-shift	$e^{at}f(t)$	$F(s - a)$
Time Scaling	$f(at)$	$\frac{1}{a}F\left(\frac{s}{a}\right)$

One-sided Laplace Transforms

f(t)	F(s)	Region of Convergence
$\delta(t)$	e^{-as} ,	whole s-plane
$u(t)$	$\frac{1}{s}$,	$\text{Re}[s] > 0$
$tu(t)$	$\frac{1}{s^2}$	$\text{Re}[s] > 0$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$	$\text{Re}[s] > 0$
$\sin(\omega t)u(t)$	$\frac{\omega}{s^2 + \omega^2}$,	$\text{Re}[s] > 0$
$\cos(\omega t)u(t)$	$\frac{2\omega_0 s}{s^2 + \omega^2}$,	$\text{Re}[s] > 0$
$t \sin(\omega_0 t)u(t)$	$\frac{(s^2 + \omega_0^2)^2}{s^2 - \omega_0^2}$,	$\text{Re}[s] > 0$
$t \cos(\omega_0 t)u(t)$	$\frac{1}{(s^2 + \omega_0^2)^2}$,	$\text{Re}[s] > 0$
$e^{-at}u(t)$	$\frac{1}{s + a}$,	$\text{Re}[s] > -a$
te^{-at}	$\frac{1}{(s + a)^2}$,	$\text{Re}[s] > -a$
$t^n e^{-at}u(t)$	$\frac{n!}{(s + a)^{n+1}}$,	$\text{Re}[s] > -a$
$e^{-\alpha t} \sin(\beta t)u(t)$	$\frac{\beta}{(s + \alpha)^2 + \beta^2}$,	$\text{Re}[s] > -\alpha$
$e^{-\alpha t} \cos(\beta t)u(t)$	$\frac{s + \alpha}{(s + \alpha)^2 + \beta^2}$,	$\text{Re}[s] > -\alpha$

Basic Properties of Fourier Series

Property	t-domain	ω -domain
Linearity	$\alpha x(t) + \beta y(t)$	$\alpha X_k + \beta Y_k$
Power	$\frac{1}{T} \int_0^T x(t) ^2 dt$	$\sum_{n=-\infty}^{\infty} X_k ^2$
Differentiation	$x'(t)$	$j k \omega_0 X_k$
Integration (only if $X_0 = 0$)	$\int_{-\infty}^t f(\tau) d\tau$	$\frac{X_k}{j k \omega_0}, k \neq 0$
Time-shifting	$x(t - \alpha)$	$e^{-j \alpha \omega_0 k} X_k$
Frequency-shift	$e^{j \alpha \omega_0 t} x(t)$	$X_{k - \alpha}, \alpha \in \mathbb{Z}$
Convolution	$z(t) = [x * y](t)$	$X_k Y_k$
Symmetry	$x(t)$ real	$ X_k = X_{-k} $ even $\angle X_k = -\angle X_{-k}$ odd

Basic Properties of CT Fourier Transforms

Property	t-domain	ω -domain
Linearity	$\alpha x(t) + \beta y(t)$	$\alpha X(\omega) + \beta Y(\omega)$
Time Scaling	$x(\alpha t), \alpha \neq 0$	$\frac{1}{ \alpha } X\left(\frac{\omega}{\alpha}\right)$
Energy	$\int_{-\infty}^{\infty} x(t) ^2 dt$	$\frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) ^2 d\omega$
Reflection	$x(-t)$	$X(-\omega)$
Duality	$X(t)$	$2\pi x(-\omega)$
Differentiation	$\frac{d^n}{dt^n} x(t)$	$(j\omega)^n X(\omega)$
Frequency Differentiation	$-jtx(t)$	$\frac{d}{d\omega} X(\omega)$
Integration	$\int_{-\infty}^t x(\tau) d\tau$	$\frac{X(\omega)}{j\omega} + \pi X(0)\delta(\omega)$
Time-shifting	$x(t - \alpha)$	$e^{-j\alpha\omega} X(\omega)$
Frequency-shift	$e^{j\omega_0 t} x(t)$	$X(\omega - \omega_0)$
Modulation	$x(t) \cos(\omega_0 t)$	$\frac{1}{2} [X(\omega - \omega_0) + X(\omega + \omega_0)]$
Periodic signals	$\sum X_k e^{j\omega_0 t}$	$\sum 2\pi X_k \delta(\omega - k\omega_0)$
Symmetry	$x(t)$ real	$ X(\omega) = X(-\omega) $ even $\angle X(\omega) = -\angle X(-\omega)$ odd
Windowing	$x(t)y(t)$	$\frac{1}{2\pi} [X * Y](\omega)$
Cosine Transform	$x(t)$ even	$\int_{-\infty}^{\infty} x(t) \cos(\omega t) dt$
Sine Transform	$x(t)$ odd	$-j \int_{-\infty}^{\infty} x(t) \sin(\omega t) dt$

CT Fourier Transform Pair

$\mathbf{x(t)}$	$\mathbf{X(\omega)}$	
1	$2\pi\delta(\omega)$	
$\delta(t)$	1,	
$\delta(t - \tau)$	$e^{-j\omega\tau}$	
$u(t)$	$\frac{1}{j\omega} + \pi\delta(\omega)$	
$u(-t)$	$\frac{-1}{j\omega} + \pi\delta(\omega)$	
$\text{sgn}(t) = 2[u(t) - 0.5]$	$\frac{2}{j\omega}$	
$e^{-at}u(t)$	$\frac{1}{a + j\omega}$,	$a > 0$
$e^{at}u(-t)$	$\frac{1}{a - j\omega}$,	$a > 0$
$te^{at}u(t)$	$\frac{1}{(a + j\omega)^2}$,	$a > 0$
$e^{-a t }$	$\frac{2a}{a^2 + \omega^2}$,	$a > 0$
$\cos(\omega_0 t)$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	
$\sin(\omega_0 t)$	$-j\pi[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$	
$u(t + \tau) - u(t - \tau)$	$2\tau\frac{\sin(\omega\tau)}{\omega\tau}$,	$\tau > 0$
$\frac{\sin(\omega_0 t)}{\pi t}$	$u(\omega + \omega_0) - u(\omega - \omega_0)$	
$\sum_{n=-\infty}^{\infty} \delta(t - nT)$	$\omega_0 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0)$,	$\omega_0 = \frac{2\pi}{T}$

Basic Properties of Z-Transforms

Property	t-domain	z-domain
Linearity	$\alpha x[n] + \beta y[t]$	$\alpha X(z) + \beta Y(z)$
Convolution	$(x * y)[n]$ $= \sum_k x[k]y[n - k]$	$X(z)Y(z)$
Time-delay (Causal)	$x[n - N]$ $(N \in \mathbb{Z})$	$z^{-N} X(z)$
Time-delay (Causal)	$x[n - N]$ $(N \in \mathbb{Z})$	$z^{-N} X(z) + \sum_{k=1}^N x[-k]z^{-N+k}$
Time reversal	$x[-n]$	$X(z^{-1})$
Multiplication by n	$nx[n]$	$-z \frac{d}{dz} X(z)$
Multiplication by n^2	$n^2 x[n]$	$z^2 \frac{d^2}{dz^2} X(z) + z \frac{d}{dz} X(z)$
Finite Difference	$x[n] - x[n - 1]$	$(1 - z^{-1})X(z) - x[-1]$
Accumulation	$\sum_{k=0}^n x[k]$	$\frac{X(z)}{1 - z^{-1}}$

One-sided Z-Transforms

$\mathbf{f[n]}$	$\mathbf{X(z)}$	Region of Convergence
$\delta[n]$	1,	whole z -plane
$u[n]$	$\frac{1}{1 - z^{-1}}$,	$ z > 1$
$nu[n]$	$\frac{z^{-1}}{(1 - z^{-1})^2}$	$ z > 1$
$n^2 u[n]$	$\frac{z^{-1}(1 + z^{-1})}{(1 - z^{-1})^3}$,	$ z > 1$
$\alpha^n u[n], \alpha < 1$	$\frac{1}{1 - \alpha z^{-1}}$,	$ z > \alpha $
$n\alpha^n u[n], \alpha < 1$	$\frac{\alpha z^{-1}}{(1 - \alpha z^{-1})^2}$,	$ z > \alpha $
$\cos(\omega_0 n)u[t]$	$\frac{1 - \cos(\omega_0)z^{-1}}{1 - 2\cos(\omega_0)z^{-1} + z^{-2}}$,	$ z > 1$
$\sin(\omega_0 n)u[t]$	$\frac{\sin(\omega_0)z^{-1}}{1 - 2\cos(\omega_0)z^{-1} + z^{-2}}$,	$ z > 1$
$\alpha^n \cos(\omega_0 n)u[t]$ $ \alpha < 1$	$\frac{1 - \alpha \cos(\omega_0)z^{-1}}{1 - 2\alpha \cos(\omega_0)z^{-1} + \alpha^2 z^{-2}}$,	$ z > \alpha $
$\alpha^n \sin(\omega_0 n)u[t]$ $ \alpha < 1$	$\frac{\alpha \sin(\omega_0)z^{-1}}{1 - 2\alpha \cos(\omega_0)z^{-1} + \alpha^2 z^{-2}}$,	$ z > \alpha $

Two-sided Z Transforms

$\mathbf{f[n]}$	$\mathbf{X(z)}$	Region of Convergence
$-u[-n - 1]$	$\frac{1}{1 - z^{-1}}$,	$ z < 1$
$-\alpha^n u[-n - 1]$	$\frac{1}{1 - \alpha z^{-1}}$,	$ z < \alpha $
$-n\alpha^n u[-n - 1]$	$\frac{1}{1 - \alpha z^{-1}} - \frac{1}{1 - \alpha^{-1}z^{-1}}$	$ \alpha < z < \frac{1}{ \alpha }$

Basic Properties of DT Fourier Transforms

Property	t-domain	Ω -domain
Z-transform	$x[n], X(z)$, $ z = 1 \in \text{ROC}$	$X(z = e^{j\Omega})$
Periodicity	$x[n]$	$X(e^{j(\Omega+2\pi k)})$
Time-shifting	$x[n - N]$	$e^{-j\Omega N} X(e^{j\Omega})$
Frequency-shift	$x[n]e^{j\Omega_0 n}$	$X(e^{j(\Omega-\Omega_0)})$
Convolution	$(x * y)[n]$	$X(e^{j\Omega})Y(e^{j\Omega})$
Multiplication	$x[n]y[n]$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\Omega-\theta)})d\theta$
Symmetry	$x[n] \in \mathbb{R}$	$ X(e^{j\Omega}) $ even function of Ω $\angle X(e^{j\Omega})$ odd function of Ω
Power	$\sum_{n=-\infty}^{\infty} x[n] ^2$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\Omega}) ^2 d\Omega$

Discrete-Time Fourier Transform (DTFT)

$\mathbf{x[n]}$	$\mathbf{X(e^{j\Omega})}$	
$\delta[n]$	1	$ \Omega < \pi$
1,	$2\pi\delta(\Omega)$	$ \Omega < \pi$
$e^{j\Omega_0 n}$	$2\pi\delta(\Omega - \Omega_0)$	$ \Omega < \pi$
$\alpha^n u[n], \alpha < 1$	$\frac{1}{(1 - \alpha e^{-j\Omega})^2}$	$ \Omega < \pi$
$\cos(\omega_0 n)u[n]$	$\pi[\delta(\Omega - \Omega_0) + \delta(\Omega + \Omega_0)]$	$ \Omega < \pi$
$\sin(\omega_0 n)u[n]$	$\frac{\pi}{j}[\delta(\Omega - \Omega_0) - \delta(\Omega + \Omega_0)]$	$ \Omega < \pi$
$\alpha^{ n }, \alpha < 1$	$\frac{1 - \alpha^2}{1 - 2\cos(\Omega) + \alpha^2}$	$ \Omega < \pi$
$u[n + \frac{N}{2}] - u[n + \frac{N}{2}]$	$\frac{\sin(\Omega(N+1)/2)}{\sin(\Omega/2)}$	$ \Omega < \pi$
$\alpha^n \cos(\omega_0 n)u[n]$	$\frac{1 - \alpha \cos(\Omega_0)e^{-j\Omega}}{1 - 2\alpha \cos(\Omega_0)e^{-j\Omega} + \alpha^2 e^{-2j\Omega}}$	
$\alpha^n \sin(\omega_0 n)u[n]$	$\frac{\alpha \sin(\Omega_0)e^{-j\Omega}}{1 - 2\alpha \cos(\Omega_0)e^{-j\Omega} + \alpha^2 e^{-2j\Omega}}$	

Linear Algebra

State-space Representation

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}, \mathbf{y} = C\mathbf{x} + D\mathbf{u}$$

Transfer function matrix

$$\mathbf{H}(s) = (sI - A)^{-1}B + D$$

Diagonalization

$$A = PDP^{-1}, \quad D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}, \quad P = \begin{bmatrix} | & | \\ \mathbf{v}_1 & \mathbf{v}_2 \\ | & | \end{bmatrix}$$

Matrix exponential

$$e^{At} = Pe^{Dt}P^{-1}, e^{Dt} = \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{bmatrix}$$

Other Identities

Geometric Series

$$\sum_{k=0}^n x^k = \frac{1 - x^{n+1}}{1 - x}, \quad \sum_{k=0}^{\infty} x^k = \frac{1}{1 - x} \quad (0 < x < 1)$$

Euler's Formula

$$e^{j\theta} = \cos \theta + j \sin \theta, \quad \cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}, \quad \sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$